

Comparing and contrasting the definitions of genetic data in Chinese and EU law

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Abstract

- The governance of genetic data reflects the distinct legal priorities of each jurisdiction, and the definition and scope of genetic data, critical for determining the application of specialized legal frameworks, vary significantly across legal systems. This article contrasts the definitional choices of genetic data in Chinese and EU law, revealing a fundamental divergence between paradigms rooted in national sovereignty versus individual rights.
- In China, genetic data lies at the intersection of biosecurity and data protection (security), resulting in the application of dual legal frameworks. The Biosecurity Law, with no EU statutory analogue, uniquely governs both physical human genetic materials and derived genetic data as sovereign national resources.
- In data protection, China's Personal Information Protection Law (PIPL) treats genetic data as sensitive only implicitly and contextually (ie when it qualifies as biometric or health-related), whereas the EU's General Data Protection Regulation (GDPR) explicitly designates all personal genetic data as a special category by virtue of its intrinsic link to fundamental rights. Moreover, driven by broader national priorities like national security and public health, China's unique Data Security Law (DSL) extends state control to non-personal, population-level genetic datasets classified as 'important data'.
- This article argues that these divergent definitional choices, stemming from irreconcilable

legal philosophies, create significant legal uncertainty and practical challenges for international genomic research, particularly for Sino-EU collaborations. It concludes that navigating this landscape requires a nuanced compliance literacy that extends beyond data protection to an understanding of the competing state and individual interests at play.

Introduction

The global expansion of genomics research has underscored the critical role of genetic data in advancing biomedical discoveries, particularly in identifying disease-associated genetic variants and addressing disparities in population representation. China, with its vast and genetically diverse population, is an active participant in this field. Its population's genetic data holds significant potential to enrich global genomic databases, inter alia offering insights into East Asian-specific genetic variants and helping to mitigate the underrepresentation of data from non-European populations in areas like drug target identification and evaluation.¹ In recent years, however, the global scientific community has grown concerned that China's strengthened regulation on human genetic resources (HGRs) may deter researchers in genomic and genetic research projects, particularly in cross-border collaborations.² This concern manifests in two principal ways.

First, China has framed the potential misuse of genetic data not solely as a threat to individual privacy but also as a matter of national sovereignty. It explicitly situates HGRs and related information within a biosecurity

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1 Robin G Walters and others, 'Genotyping and Population Characteristics of the China Kadoorie Biobank' (2023) 3 Cell Genomics 100361 <<https://doi.org/10.1016/j.xgen.2023.100361>> accessed 8 October 2025.

2 Dyani Lewis, 'China's Souped-up Data Privacy Laws Deter Researchers' (Nature, 25 May 2023) <<https://www.nature.com/articles/d41586-023-01638-1>> accessed 8 October 2025.

paradigm, treating them as a strategic national resource tied to biological sovereignty, public health, and national security. That legal framing has produced a tier of regulation, without a direct European statutory analogue, which non-Chinese researchers must take into account when dealing with Chinese genetic materials or data. Such divergent regulatory conceptualizations could have concrete implications for Sino-EU research collaborations with genetic data. Whereas European researchers predominantly navigate a unified General Data Protection Regulation (GDPR) framework focused on individual privacy,³ projects involving Chinese HGRs frequently entail prior governmental approvals under biosecurity procedures, heightening complexity and potential uncertainty for international consortia. This additional compliance burden risks stifling collaborative innovation unless stakeholders understand and bridge these regulatory asymmetries.

Secondly, China's data protection architecture is organized differently from that of the EU. Differences in legal hierarchy, the binding nature of complementary legal instruments, and the coexistence of multiple applicable regimes produce distinct compliance profiles. Notably, unlike the GDPR, China's data protection law appears to treat individuals' genetic data as sensitive only indirectly. Under China's unique framework for non-personal data governance, population-level genetic datasets, in certain circumstances, are classified as 'important data', a less precisely defined legal category that attracts heightened data-security requirements. The legally defined scope of genetic data therefore underpins other related regulatory questions, determining the intensity of the regulatory burden that those processing such data must endure.

Against this backdrop, this article aims to explore how Chinese and EU legal instruments define and delimit genetic data and related terms, and how these differences may produce practical consequences for researchers using genetic data, especially those in international collaborations. Methodologically, it deploys a hybrid doctrinal and comparative approach. The doctrinal component consists of close textual analysis and purposive interpretation of primary legal instruments and legally binding rules, supplemented by a targeted review of non-binding regulatory guidance, national standards, and administrative instruments where available. The comparative component systematically maps two jurisdictions against a set of evaluative dimensions—mainly including the legal positioning of genetic data, formal definitions, and sensitivity classification—to translate abstract definitions into operational

consequences. For transparency and practitioner utility, this article also provides a consolidated crosswalk of terms as an Appendix [Table 1](#), thereby enabling side-by-side comparison and facilitating application to real-world research scenarios.

The remainder of this article proceeds as follows. The section 'The Chinese legal system: a distinct context for genetic data regulation' sets out the background of the distinct Chinese legal system which is necessary for understanding genetic data regulation in China. The section 'China's dual frameworks governing the concept of genetic data: national sovereignty versus individual rights' identifies China's dual legal frameworks governing the definition(s) of genetic data—biosecurity and data protection—and their competing paradigms of national sovereignty versus individual rights. The section 'Genetic data as a national resource: the wide reach under the biosecurity law' then delves into the first, analysing how the biosecurity framework conceptualizes genetic data as a strategic national resource. The section 'Genetic data under the Chinese data protection law' examines the second, detailing the dual status of genetic data as both sensitive personal data and, at the population level, as non-personal 'important data'. Building on this foundation, the section 'Comparative analysis: divergent definitional choices and paradigms in Chinese and EU law' provides the central comparative analysis, systematically contrasting the Chinese and EU legal frameworks across key definitional and structural divergences. The concluding section synthesizes this article's findings and briefly discusses their practical implications for researchers engaging in international genomic collaboration.

As the primary contribution of this article, it identifies the similarities and differences in the definitions of genetic data provided by these two jurisdictions, both in *de jure* and *de facto* terms, and how the diverging definitional choices may influence researchers using genetic data. These implications are also shaped by the differing nature between China's guidance-type regulatory documents and the EU's data protection guidelines, which are commonly used in the rapidly evolving field of genetic research. It should be noted that publicly available judicial decisions and formal regulatory determinations addressing the definition and scope of genetic data remain limited, particularly in China. Consequently, this article principally relies upon statutory texts, implementing regulations, national standards, representative regulatory guidance, and publicly reported administrative notices as the evidentiary basis for its analysis. The absence of readily accessible, authoritative enforcement

3 Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard

to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC GDPR, OJ 2016 L 119/1.

Table 1. Comparative crosswalk of key legal definitions referenced in this article.

Definitions in Chinese law		Definitions in EU law	
Term	Definition and source	Term	Definition and source
Sensitive personal information	Personal information that, once leaked or illegally used, may easily cause harm to the dignity of natural persons grave harm to personal or property security, including information on biometric characteristics, religious beliefs, specially designated status, medical health, financial accounts, individual location tracking, etc, as well as the personal information of minors under the age of 14 years. (Article 28 PIPL)	Sensitive personal data	Including personal data revealing racial or ethnic origin, political opinions, religious or philosophical beliefs, or trade union membership, and the processing of genetic data, biometric data for the purpose of uniquely identifying a natural person, data concerning health or data concerning a natural person's sex life or sexual orientation. (Article 9(1) GDPR)
HGR	Including both HGR materials and HGR information. (Article 85(8) Biosecurity Law and Article 2 RMHGR)		
HGR material	Genetic materials such as organs, tissues, and cells, that contain the human genome, genes, and other genetic materials. (Article 85(8) Biosecurity Law and Article 2 RMHGR)		
HGR information	Data and other information materials produced from HGR materials. (Article 85(8) Biosecurity Law and Article 2 RMHGR) Information materials such as data on human genes and genomes produced using HGR materials ... not including clinical data, imaging data, protein data, and metabolic data. (Article 2 RMHGR Implementation Rules)	Genetic data	Personal data relating to the inherited or acquired genetic characteristics of a natural person which give unique information about the physiology or the health of that natural person and which result, in particular, from an analysis of a biological sample from the natural person in question. (Article 4(13) GDPR)
Genetic recognition data	Data representing the genetic information of an individual or group extracted from human genetic material through technical means, which can directly or indirectly identify a human individual or group. (National standard GB/T		

Continued

Table 1. *Continued*

Definitions in Chinese law		Definitions in EU law	
Term	Definition and source	Term	Definition and source
Important data	41806-2022) (DSL, Regulations on the Management of Network Data Security)	/	
Biometric recognition information	Personal information obtained through technical processing of the physical, biological, or behavioural characteristics of a natural person, which can independently or in combination with other information be used to identify the identity of that natural person. This includes personal facial recognition features, iris, fingerprints, genes, voiceprints, gait, palm prints, earlobes, eye patterns, etc. (National standard GB/T 40660-2021)	Biometric data	Personal data resulting from specific technical processing relating to the physical, physiological, or behavioural characteristics of a natural person, which allow or confirm the unique identification of that natural person, such as facial images or dactyloscopic data. (Article 4(14) GDPR)
Personal health data	Electronic data that, when alone or in combination with other information, can identify a specific natural person or reflect the physical or mental health of a specific natural person. This includes (but is not limited to): (a) data results from examinations or tests of body parts or body substances; (b) medical research-related data, such as clinical research case data, biobanks, various bioinformatics sequencing results (including whole genome sequencing), etc. (National standard GB/T 39725-2020)	Health data (data concerning health)	Personal data related to the physical or mental health of a natural person, including the provision of health care services, which reveal information about his or her health status. (Article 4(15) GDPR)

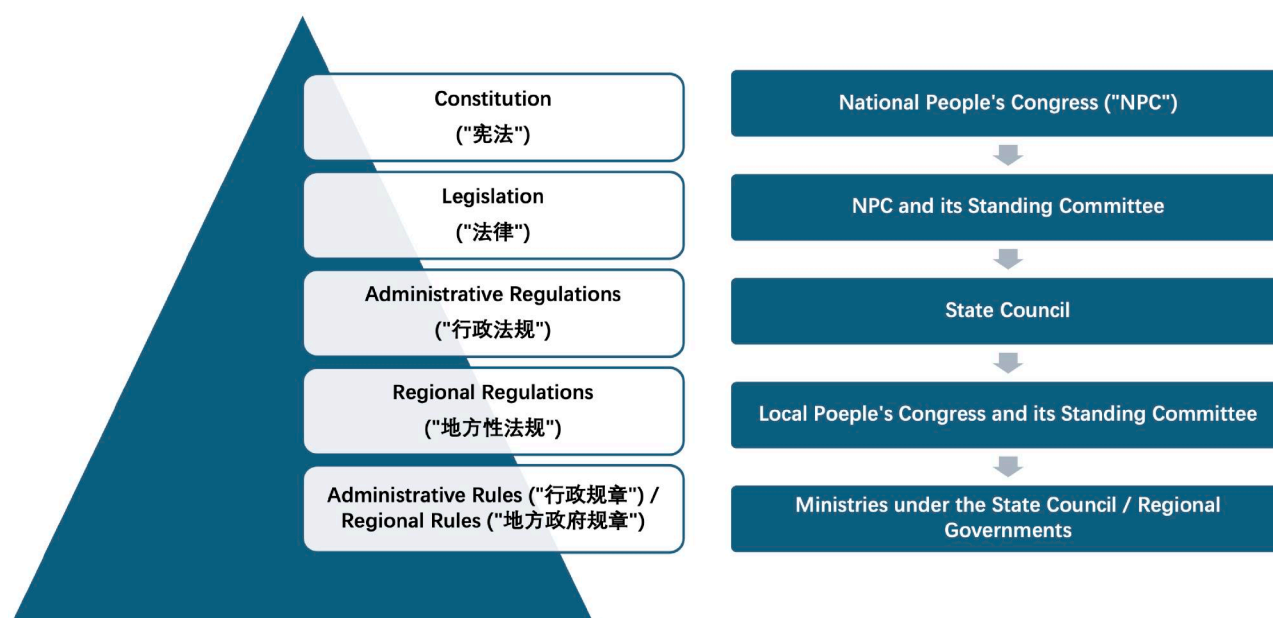


Figure 1. Hierarchical structure of the Chinese legal system.

cases is therefore an important caveat to the practical inferences offered herein.

The Chinese legal system: a distinct context for genetic data regulation

There are important differences between how the Chinese legal system operates and how the legal regimes of both the EU and its Member States function, which influence the regulation of genetic data. China's legal system, deeply rooted in its particular political system where the Constitution⁴ guarantees the Communist Party's supreme authority,⁵ absorbs the Civil Law traditions and is influenced by a socialist legal system while maintaining 'Chinese characteristics'.⁶ Laid down in the 'Legislation Law',⁷ the hierarchy of the Chinese legal system explains the relationship among different types of legislation, regulations, and rules (hereinafter collectively 'laws') at the central level, as well as the vertical relationship between central-level law and regional-level law. The Legislation Law also fundamentally establishes

the legislative powers of different state agencies and law-making procedures for China's rule of law. Figure 1 shows an overview of this hierarchy and the respective enacting institutions within China's political system. As the section 'Conclusion' will discuss, the particular hierarchical structure of the Chinese legal system may have important implications when interpreting the application of certain laws relevant to genetic research.

At the central level, the Constitution has the highest legal authority, and the development of all types of legislation shall not contradict the Constitution.⁸ The National People's Congress (NPC) and its Standing Committee exercise their legislative power to enact nationwide applicable legislation,⁹ including basic legislation and other national legislation.¹⁰ The statutory position of legislation in China is arguably similar to the primary legislation in the Member States' law and harmonized regulations in the EU law (eg the EU's GDPR).

Based on legislation, the State Council, China's central government in Beijing, has the power to formulate administrative regulations in order to implement

4 Constitution of the People's Republic of China (中华人民共和国宪法) (promulgated by the National People's Congress, 4 December 1982, lastly amended 11 March 2018) ('PRC Constitution').

5 Lingqiao Song and Rosario Isasi, 'The Regulation of Human Germline Genome Modification in the People's Republic of China' in Andrea Boggio, Cesare PR Romano and Jessica Almqvist (eds), *Human Germline Genome Modification and the Right to Science* (1st edn, CUP 2020) <https://www.cambridge.org/core/product/identifier/9781108759083%23CN-bp-17/type/book_part> accessed 9 October 2025.

6 Mo Zhang, 'The Socialist Legal System with Chinese Characteristics: China's Discourse for the Rule of Law and a Bitter Experience' (2011) 24 *Temple International & Comparative Law Journal* 1.

7 Legislation Law of the People's Republic of China (中华人民共和国立法法) (promulgated by the National People's Congress, 15 March 2000, lastly amended 13 March 2023) ('PRC Legislation Law').

8 PRC Constitution, art 5.

9 *ibid*, art 59.

10 PRC Legislation Law, art 10. The basic legislation is enacted by the NPC, including criminal laws, civil laws, and laws governing state organs. Other national legislation is enacted and issued by the Standing Committee of the NPC, covering various sectors such as data protection, cybersecurity, biosecurity, and consumer protection.

provisions in legislation and discharge its administrative duties provided by the Constitution.¹¹ Although these administrative regulations have a lower hierarchical value than legislation, they are nonetheless of a legally binding nature. Such laws are arguably similar in nature to forms of secondary law enacted throughout much of Europe. For example, the Regulation on the Management of Human Genetic Resources (as discussed further in the section ‘Comparative analysis: divergent definitional choices and paradigms in Chinese and EU law’) plays an important role in the regulatory regime for the use of genetic data in China, which is an administrative regulation enacted by the State Council in the hierarchy of the Chinese legal system. A level lower, the ministries of the State Council have the power to develop departmental rules within their own mandates to specify matters outlined in these upper-level laws.¹² Such laws are arguably similar to ministerial decrees that exist in various countries in Europe. In China, in this regard, the ministries of health and science and technology can accordingly develop detailed rules for the use of genetic data due to their regulatory competence in this field.

At the regional level, the regional people’s congress and its standing committee of provinces and municipalities may develop regional regulations to tailor national-level laws to local needs, provided that they do not contradict higher-level laws.¹³ If regional regulations create an important divergence from national-level laws, for example, where they are developed to address gaps not covered by higher-level laws, they may require modification or repeal to align with central-level legislation or administrative regulations.¹⁴ Regional governments (ranging from provinces to municipalities) may also develop regional rules to implement upper-level laws in light of local particular contexts.¹⁵ These binding regional rules, which share the same hierarchical level as departmental rules developed by central ministries, can introduce further variability in implementation and interpretation, necessitating central review and harmonization by the State Council.¹⁶ Given the excessive number of regional laws in China, this article will focus

on China’s central level laws towards the comparative discussion of genetic data definition.

China’s dual frameworks governing the concept of genetic data: national sovereignty versus individual rights

The governance of genetic data in China is uniquely characterized by a dual-track legislative approach, reflecting a tension between two distinct paradigms: the safeguarding of national sovereignty and the protection of individual rights. This section introduces the two principal legislative frameworks that embody this duality: one is the Biosecurity Law and another is the Personal Information Protection Law (PIPL), and outlines their fundamentally different philosophical underpinnings. This will be essential for understanding the detailed in-depth analysis in subsequent sections.

The Biosecurity Law: a paradigm of national sovereignty

China’s Biosecurity Law, effective from 15 April 2021,¹⁷ is the first piece of comprehensive and systematic legislation that fills the absence of a basic law in the biosecurity field in China.¹⁸ It contains inter alia a chapter on the security of HGRs and biological resources,¹⁹ under which genetic data (ie sequenced data) is subsumed. The law therefore covers two related but separate areas: the creation and use of genomic data, and the collection and use of biological materials from which they can be derived.

Notably, the Biosecurity Law was not the first legislative initiative in China concerned with the regulation of genetic data. China’s legislative activities in the field of HGRs can be traced back to the 1990s, when the State Council of China issued the Interim Measures on the Management of Human Genetic Resources on 10 June 1998, the earliest legislative effort in regulating genetic materials and data of human beings.²⁰ It was repealed and replaced by the Regulation on the Management of Human Genetic Resources (RMHGR) in 2019 (to be

11 PRC Legislation Law, art 72.

12 *ibid*, art 91.

13 *ibid*, art 80.

14 *ibid*, art 82.

15 *ibid*, art 93.

16 *ibid*, art 106. The Legislation Law does not explicitly address the hierarchy relationship between regional regulations formulated by regional people’s congress and departmental rules developed by ministries of the State Council. The Legislation Law leaves the decision-making power to the State Council and the Standing Committee of NPC in case of any inconsistency between these two different types of laws on a specific matter.

17 Biosecurity Law of the People’s Republic of China (中华人民共和国生物安全法) (promulgated by the Standing Committee of the National People’s Congress, 17 October 2020, effective 15 April 2021) (‘PRC Biosecurity Law’).

18 Liang Huigang and others, ‘Significance of and Outlook for the Biosecurity Law of the People’s Republic of China’ (2021) 3 *Journal of Biosafety and Biosecurity* 46.

19 PRC Biosecurity Law, Ch VI.

20 Interim Measures on the Management of Human Genetic Resources (人类遗传资源管理暂行办法) (promulgated by the State Council, 10 June 1998, abolished 1 July 2019) <<https://www.cncbd.org.cn/News/Detail/3374>> accessed 9 October 2025.

discussed in the section ‘Comparative analysis: divergent definitional choices and paradigms in Chinese and EU law’). In history, two well-known developments substantially accelerated China’s HGR-related legislative activities. The first was the wave of biopiracy by foreign-funded researchers in the late 1990s, which was called ‘Gene War of the Century’ by human genetics researchers.²¹ This trend encouraged China to give the management of HGRs its own dedicated legislative regime. This was also provoked by the progression of scientific research in the 1990s, together with the realization of the vast potential that China, having a very large population, would offer in terms of the scale of genetic data that could be made available to researchers.²² It can also arguably be attributed to a change in the global geopolitical environment.²³ The second was the scandal of gene-edited babies in 2018, which brought unwanted negative foreign attention to Chinese research practices, resulting in much pressure to improve its laws and ethical norms relating to genetic research.²⁴ With the adoption of the Biosecurity Law in 2020, China finally established a formal basis for regulating HGRs and the legal status of genetic data as a significant focus for legislation.

At its heart, the Biosecurity Law is anchored in a collectivist and nation-oriented philosophy that has no direct equivalent in EU law. The law itself bears some similarities to initiatives elsewhere, both on the national and international level, including the UN Convention on Biological Diversity²⁵ and jurisdictions such as Indonesia that had (ie before China) recently created their own laws on bio-sovereignty.²⁶ These national laws emerged against the backdrop of colonial and post-colonial imbalances, where developing countries often perceived systematic exploitation in bioscience and pharmacy. For instance, breakthroughs based on biological materials from developing countries routinely did

not lead to fair remuneration for the source countries.²⁷ Such extractive activities, typically driven by established Western companies or research institutions, have been described as ‘bioprospecting’ or, in their most egregious form, ‘biopiracy’.²⁸ Given that most Western countries have not faced comparable issues, it is hardly surprising that the EU has not enacted equivalent legislation.

While some other developing nations have embraced a similar logic,²⁹ no parallel exists under EU law or within its Member States’ laws concerning genetic data. China’s Biosecurity Law, however, goes much further, most notably because it covers not only HGRs but also those from non-human organisms.³⁰ Its domain of application thus far exceeds that of most other equivalents in developing countries and even the UN Convention on Biological Diversity. Consequently, its provisions on human genetic data are anchored in the recognition that Chinese people and the nation form a separate and distinguishable group whose collective biological assets require state protection. It explicitly situates HGRs within a biosecurity paradigm, treating them as a strategic national asset intrinsically linked to public health and national security.³¹

This sovereignty-centric rationale manifests in strict regulatory controls. Under the Biosecurity Law, activities involving China’s HGRs, including the collection, preservation, utilization for international scientific collaboration, and external provision, must be subject to an ex ante government approval procedure.³² When providing or granting access to genetic information to foreign organizations or individuals, there are also prior obligations of reporting to the competent regulator and submitting a backup of the information.³³ For genetic research projects with an international collaboration aspect, obstacles may arise not only before the research but also during the research. For example, the regulator may sideline scientific research projects during the

21 Sun-Wei Guo, Chang-Jiang Zheng and CC Li, ‘“Gene War of the Century?”’ (1997) 278 *Science* 1693.

22 Joy Y Zhang, ‘What Do the British and Chinese Governing Visions on Human Genomic Research Tell Us about Biosovereignty?’ (2023) 25 *Asian Biotechnology and Development Review* 5.

23 Sun-Wei Guo, ‘China’s “Gene War of the Century” and Its Aftermath: The Contest Goes On’ (2013) 51 *Minerva* 485.

24 Lingqiao Song and Yann Joly, ‘After He Jianku: China’s Biotechnology Regulation Reforms’ (2021) 21 *Medical Law International* 174.

25 For a recent discussion of attempts to reform the framework created by the convention, see: Elsa Tsioumani, ‘Convention on Biological Diversity: A Review of the Post-2020 Global Biodiversity Framework Working Group Negotiations’ (2020) 50 *Environmental Policy and Law* 55.

26 Zhang (n 22); Catherine Rhodes, ‘Potential International Approaches to Ownership/Control of Human Genetic Resources’ (2016) 24 *Health Care Analysis* 260.

27 In Indonesia, eg this included during an Influenza outbreak where pharmaceutical companies use genetic samples (of non-human origin) to help in their understanding of the virus. See: Rhodes (n 26).

28 Eva Hilberg, ‘Molecular Sovereignities: Patients, Genomes, and the Enduring Biocoloniality of Intellectual Property’ (2022) 17 *Biosocieties* 695.

29 Béatrice Séguin and others, ‘Genomic Medicine and Developing Countries: Creating a Room of Their Own’ (2008) 9 *Nature Reviews Genetics* 487; Zhang (n 22).

30 Smriti Mallapaty, ‘China Expands Control over Genetic Data Used in Scientific Research’ (2022) 605 *Nature* 405.

31 Kyle McKibbin and Mahsa Shabani, ‘Genomic Data as a National Strategic Resource: Implications for the Genomic Commons and International Data Sharing for Biomedical Research and Innovation’ (2023) 51 *Journal of Law, Medicine & Ethics* 301.

32 According to art 56 of the PRC Biosecurity Law, the collection activities only refer to those collecting HGRs from important genetic lineages, specific regions of China, or other types and quantities as specified by the competent department of science and technology under the State Council.

33 PRC Biosecurity Law, art 57.

review process if they involve the collection of samples and information from a genetically distinct population.³⁴ Even if the international collaboration research project has been permitted, the regulator still retains statutory power to revoke approved licenses based on concerns about national security, state competitiveness, and public health.³⁵

The personal information protection law

In parallel with the biosecurity governance regime, genetic data also receives special attention under China's data protection framework. The PIPL, effective from 1 November 2021 as one of the pillars of China's data protection regime,³⁶ applies to the processing activities of personal information of individuals within China and extends its reach extraterritorially in specified circumstances.³⁷ While the PIPL draws certain structural and substantive inspirations from the EU's GDPR—for instance, by distinguishing between general and sensitive personal information and imposing stricter requirements for the latter (although there are important differences between the two in how genetic data is portrayed)—it is fundamentally shaped by China's unique cultural values, legal traditions, and political situation.³⁸

The underlying values of the PIPL reflect a balancing act between protecting individuals' information rights and advancing broader state objectives, including public interest, economic development, and national data security.³⁹ This contrasts sharply with similar data protection initiatives in Europe, like the GDPR, whose ethos is more singularly focused on privacy as a fundamental human right.⁴⁰ Although both laws function to regulate activities of processing personal data, the PIPL operates within a framework of significant state oversight and in parallel with another governing framework with a completely different legislative philosophy, leading to important divergences in how genetic data is conceptualized and protected. Whereas the EU framework is

designed primarily to safeguard individual autonomy and dignity, the PIPL situates these protections within a broader context of state interests.

Thus, an individual's genetic data primarily enjoys dual legal identity in Chinese law: being HGRs under the Biosecurity Law as a national asset, and being a form of (sensitive) personal information under the PIPL. Understanding such a dual legal identity—where genetic data is at once a matter of national sovereignty under one law and a matter of individual rights under another—is critical to navigating China's complex regulatory landscape. The subsequent sections of this article will analyse each framework in greater depth before comparing their practical implications with the EU's approach.

Genetic data as a national resource: the wide reach under the biosecurity law

Broad scope, conceptual ambiguities, and potential overreach

The Biosecurity Law expressly announces the sovereignty China enjoys over HGRs of the country,⁴¹ covering both 'human genetic resource materials' ('人类遗传资源材料', HGR materials) and 'human genetic resource information' ('人类遗传资源信息', HGR information), two important concepts defining the scope of genetic data through the biosecurity lens. Under Article 85(8) of the Biosecurity Law, the former refers to physical 'genetic materials, including organs, tissues, and cells, that contain the human genomes, genes and other genetic substances', and the latter relates to the data and other information materials sequenced from HGR materials.⁴² Instead of the expression of 'genetic data', the Biosecurity Law uses the term 'HGR information' to refer to the electronic form. These two important definitions mirror, in a nearly identical manner, those promulgated two years earlier in the RMHGR,⁴³ an

34 Patrick Beyrer, 'China's Securitization of Genetic Research' (*The Diplomat*, 13 March 2023) <<https://thediplomat.com/2023/03/chinas-securitization-of-genetic-research/>> accessed 9 October 2025.

35 Yongxi Chen and Lingqiao Song, 'China: Concurring Regulation of Cross-Border Genomic Data Sharing for Statist Control and Individual Protection' (2018) 137 *Human Genetics* 605.

36 Rogier Creemers, 'China's Emerging Data Protection Framework' (2022) 8 *Journal of Cybersecurity* tyac011.

37 Personal Information Protection Law of the People's Republic of China (《中华人民共和国个人信息保护法》) (promulgated by the Standing Committee of the National People's Congress, 20 August 2021, effective 1 November 2021) ('PIPL').

38 Wenlong Li and Jiahong Chen, 'From Brussels Effect to Gravity Assists: Understanding the Evolution of the GDPR-Inspired Personal Information Protection Law in China' (2024) 54 *Computer Law & Security Review* 105994.

39 PIPL, art 1.

40 Li and Chen (n 38).

41 PRC Biosecurity Law, art 53. Notably, the Biosecurity Law extends the sovereignty approach to biological resources in China as well, that is animal and plant resources. According to art 2, except for regulating HGRs and biological resources in China, this law also applies to other activities related to biological security, including the prevention and control of major emergent infectious diseases and outbreaks in animals and plants, biotechnology R&D, biological security management in pathogenic microbiology laboratories, preventing encroachment by foreign species and the preservation of biodiversity, responding to microbial drug resistance, and preventing bioterrorist attacks and defending against the threat of biological weapons.

42 PRC Biosecurity Law, art 85(8).

43 Regulation on the Management of Human Genetic Resources (《人类遗传资源管理条例》) (promulgated by the State Council of the People's Republic of China, 20 March 2019, effective 1 July 2019) ('RMHGR').

administrative regulation adopted by China's State Council in 2019 before the Biosecurity Law. The RMHGR sought chiefly to prevent the illegal and unethical use of HGRs in biotechnological research activities, particularly in international research collaboration using Chinese HGRs,⁴⁴ including more detailed and targeted rules than the Biosecurity Law. The RMHGR remains effective after the promulgation of the Biosecurity Law, and in cases of tension and inconsistency, however, the higher hierarchical status of the more recent Biosecurity Law prevails, thereby enhancing legal certainty and governance coherence.

The breadth of the statutory definitions gives rise to conceptual ambiguities, particularly the phrase 'information produced from HGR materials' which is not precisely bounded. A literal and semantic reading could suggest that any information produced from HGR materials falls into the legal scope of genetic data in the Biosecurity Law and the RMHGR. This would seemingly imply a very broad scope of what can constitute genetic data (eg going far beyond the concept in the GDPR). Particularly, it does not explicitly state that such information should reveal the heritable characteristics of individuals, potentially creating a very broad scope of application to all types of data generated from human biological samples. The inclusion of catch-all terms like 'and other information materials' also suggests a broad catchment. This vagueness raises important concerns surrounding the regulatory boundary of 'HGR information',⁴⁵ potentially creating issues in terms of legal certainty, something that could have a deterrent effect on potential research, not only in areas that clearly involve genetic research but also those that do not.⁴⁶

A good example revealing the tension within China's definition of genetic data is the case of blood tests,

which can be used in various ways and for different purposes, such as extracting DNA samples for screening certain genetic conditions or conducting full blood count (FBC) tests to assess general health risks.⁴⁷ Whereas next-generation sequencing of genomic loci or panels indisputably yields highly sensitive personal data, facilitating inferences about ancestry, disease risk, or pharmacogenomic responses, routine FBC parameters do not disclose genetic sequences or heritable characteristics in the same sense. The key distinction is therefore one of informational granularity and inferential power. Under the Biosecurity Law and the RMHGR, however, both outputs derive from 'HGR materials' and thus are subsumed under the same legal scope of 'HGR information', so that low-granularity clinical outputs and high-granularity genomic sequences can fall within the same regulatory category. The authors therefore argue that labelling both types of data as human genetic information is problematic, given their innate difference in sensitivity: high-resolution genetic data can reveal highly sensitive individual, familial, and group attributes, and its scope of revelation is expanding with technological advances.⁴⁸ It can therefore be argued that the statutory framing in China's Biosecurity Law and the RMHGR, by encompassing all information derived from HGRs, produces a regulatory ambit that is both too vague and broad.

Clarifications in the 2023 implementation rules and regulatory guidance

In June 2023, the Ministry of Science and Technology ('MoST'), the competent central HGR regulator at the time,⁴⁹ issued the binding departmental rules Implementation Rules for the Regulations on the Management of Human Genetic Resources ('RMHGR Implementation Rules') to refine the regime.⁵⁰ This

44 Ministry of Science and Technology of the People's Republic of China, 'Officials from the Ministry of Justice and the Ministry of Science and Technology answered questions on the Regulation of the People's Republic of China on the Management of Human Genetic Resources (司法部、科技部负责人就《中华人民共和国人类遗传资源管理条例》答记者问)' (12 June 2019) <https://www.most.gov.cn/xxgk/xinxi/fenlei/fdzdgknr/fgzcz/cjd/202106/t20210625_175392.html> accessed 8 October 2025.

45 Richard Bartlett, 'Human Genetic Resources' (*King & Wood Mallets*, 18 November 2020) <<https://www.chinalawinsight.com/2020/11/articles/compliance/human-genetic-resources/>> accessed 8 October 2025.

46 Chassang Gauthier, 'The Impact of the EU General Data Protection Regulation on Scientific Research' (2017) 11 *Eancermediscience* 709.

47 National Health System of the United Kingdom, 'Blood tests' <<https://www.nhs.uk/conditions/blood-tests/types/>> accessed 8 October 2025.

48 Paul Quinn and Liam Quinn, 'Big Genetic Data and Its Big Data Protection Challenges' (2018) 34 *Computer Law & Security Review* 1000.

49 The MoST is an executive department of the State Council of China, responsible for coordinating activities and developing regulations and policies regarding science and technology within the country. It has been the

main competent regulator in terms of managing affairs on HGRs for a long time, either in name or in substance, before the institutional reform of the State Council in 2023. According to the Reform Plan for the Party and State Institutions released by the Communist Party of China and the State Council on 16 March 2023, the National Health Commission would be responsible for administrative affairs of HGRs instead of the MoST in the future. This shift of competent regulatory authority finally took into effect since 1 May 2024. See Central Committee of the Communist Party of China and the State Council, 'Plan on Reform of Party and State Institutions' (党和国家机构改革方案) (16 March 2023) <https://www.gov.cn/zhengce/2023-03/16/content_5747072.htm> accessed 8 October 2025; State Council of the People's Republic of China, 'Decision of the State Council on Amending and Repealing Certain Administrative Regulations' (国务院关于修改和废止部分行政法规的决定) (15 March 2024) <https://www.gov.cn/zhengce/content/202403/content_6939590.htm?ddtab=true> accessed 8 October 2025.

50 Implementation Rules for the Regulations on the Management of Human Genetic Resources (人类遗传资源管理条例实施细则) (promulgated by the MoST, effective 1 July 2023) ('RMHGR Implementation Rules').

issuance is significant for clarifying the regulatory boundary of HGRs, particularly those in digital form. Most notably, it narrows the Biosecurity Law's broad concept of 'HGR information' by defining it as information 'concerning human genes and genomes generated from 'HGR materials' and expressly excluding certain data types like clinical data, image data, protein data and metabolic data'.⁵¹ This is a welcome clarification to the nature of the definition of 'HGR information' as provided in the Biosecurity Law and the RMHGR, since it narrows the scope to human gene and genomic data that are generated from physical samples of HGRs, rather than all types of information that could be derived from them. This clarification presumably enables researchers to adopt a more confident compliance approach, as the biosecurity governance requirements now clearly apply to raw genetic data and other genomic data generated from physical HGR samples.⁵²

However, given the Biosecurity Law's superior position in the Chinese legal hierarchy, the exclusion of these data types cannot be considered absolute. This refined definition should not be interpreted rigidly, especially in light of advancing genomic applications. For example, in clinical care settings, genetic testing can provide information on drug effectiveness, which enables individualized drug doses or helps identify a patient's risk of severe adverse events, facilitating better medical treatment decisions.⁵³ This is a kind of diagnostic testing methodology in the clinical setting, and genomic test results are often integrated into a patient's electronic health record (EHR).⁵⁴ In such cases, the EHR, a form of clinical data, contains certain genetic information that may reveal heritable characteristics of the patient. Considering the MoST's clarifying purpose, it may still be necessary to assess whether such data is generated from 'HGR materials' of individuals in China and contains any information about human genes or genomes, which could bring it back within the superior Biosecurity Law's purview.

Subsequently, regulatory guidance further refined the scope of 'HGR materials' by exempting human secretions, body fluids, and swabs that do not contain cells.⁵⁵ In September 2023, the MoST further narrowed the actual scope by excluding 'biological samples such as urine, feces, serum and plasma, which may contain a very small amount of shed, residual or free cells or genes'.⁵⁶ In other words, this exemption acknowledges the disproportionate difficulty of sequencing from such biological materials due to the minute existence of human genetic material they hold. In parallel, the regulator clarified that 'HGR information' affirmatively includes data concerning human genes, genome, transcriptome, epigenome, ctDNA, and other nucleic acid biomarkers, as well as related association data (eg disease and race).⁵⁷ These clarifications, either directly or indirectly, continue to shape the scope of genetic data under China's biosecurity framework.

Despite these exemptions for certain biological materials, China's biosecurity regulator does not treat the prospect of deriving genetic data from these samples as impossible, especially when scientific research employs assays for genes, genome, transcriptome, epigenome, and nucleic acid biomarkers.⁵⁸ The regulatory position is that information generated from these 'exempt' samples remains subject to the biosecurity regime in specific high-risk contexts, namely international scientific collaboration, the external provision of such information, or its open use. These activities still require administrative approval, filing, or advanced reporting to the biosecurity regulator.⁵⁹ In short, the exemption of specific sample types does not create a blanket exclusion in the research context. This demonstrates that scientific research activities receive particular scrutiny due to their potential to produce high-resolution, shareable genetic datasets, involve international collaboration and cross-border transfers, and thus implicate public-health, biosecurity, and national-sovereignty concerns. A notable illustration is a publication of a Chinese pangenome study in an international academic journal in 2023,⁶⁰

51 RMHGR Implementation Rules, art 2.

52 Luopei Wei and others, 'Policy Interpretation of the Implementation Rules of the Regulations on the Management of Human Genetic Resources and Analysis of the Management Strategies in the New Era' (2024) 37 Chinese Journal of Medical Science Research Management 98.

53 Nora Franceschini, Amber Frick and Jeffrey B Kopp, 'Genetic Testing in Clinical Settings' (2018) 72 American Journal of Kidney Diseases 569.

54 Theresa A Grebe and others, 'The Interface of Genomic Information with the Electronic Health Record: A Points to Consider Statement of the American College of Medical Genetics and Genomics (ACMG)' (2020) 22 Genetics in Medicine 1431.

55 MoST, 'The Notice on Updating the Service Guide, Filing and Prior Reporting Scope and Procedures for Administrative Licensing Matters of Human Genetic Resources (关于更新人类遗传资源行政许可事项服务指南、备案以及事先报告范围和程序的通知)' (14 July 2023) <www.most.gov.cn/tztg/202307/t20230714_187075.html> accessed 8 October 2025.

56 MoST, 'Notice on Publishing FAQs on Human Genetic Resources Management (关于发布人类遗传资源管理常见问题解答的通知)' (12 September 2023) <https://www.most.gov.cn/tztg/202309/t20230912_187931.html> accessed 8 October 2025.

57 MoST (n 55).

58 MoST (n 56).

59 *ibid.*

60 In June 2023, a group of researchers from Fudan University published an article entitled *A pangenome reference of 36 Chinese populations* on *Nature* (DOI: <<https://doi.org/10.1038/s41586-023-06173-7>>), which analyses genomic sequences from 58 individuals representing 36 minority ethnic groups in China and captures genomic variability that is not currently reflected in human reference genomes.

which may constitute an ‘external provision’ or ‘open use’ under China’s biosecurity regime. Public statements associated with that publication expressly indicated that the authors sought and received prior approval from the HGR regulator before internationally publishing their findings,⁶¹ although the formal approval criteria or the regulator’s detailed reasoning applied in that instance remain unavailable in the public record.

Ultimately, these clarifications suggest the regulator’s focus is on the intention to sequence genetic data in the definition of ‘HGR information’. The targeted exemptions of sample materials signal that China does not aim to develop an entirely risk-free approach but instead seeks to mitigate biosecurity risks to a relatively acceptable level. While such regulatory interpretation does not carry the same legal effect as the Biosecurity Law itself, it still provides vital guidance for understanding the real-world scope of genetic data and compliance practices under this framework.

Genetic data under the Chinese data protection law

Genetic data as sensitive personal information

China’s PIPL distinguishes ‘sensitive personal information’ from general personal information on the grounds that its misuse ‘may easily cause harm to the dignity, personal security, or property security of individuals’.⁶² While the PIPL does not explicitly list genetic data, information on biometric characteristics, and medical health of a natural person are two designated categories of ‘sensitive personal information’ under the PIPL.⁶³ This section argues that the PIPL implicitly includes various forms of genetic data as sensitive through these two categories, a contrast to the EU’s GDPR, which

comprehensively classifies all forms of human genetic data as a special category *per se*.⁶⁴

In practice, genetic data used to uniquely identify an individual based on permanent physiological characteristics will likely fall within the scope of biometric data.⁶⁵ Similarly, genetic data that can reveal the health status of an individual (eg data from genetic disease screening), or can be simply compared to a disease-specific genomic database with some degree of public accessibility for ‘beacon’ style queries, will likely be considered personal medical health data.⁶⁶ The fact that the PIPL does not explicitly treat all genetic data as *per se* sensitive data, however, is not without significance. For example, genetic data generated in blue-sky scientific research or other research with non-health-related goals (eg studies of human evolution or migration patterns) would not, on its face, reveal individual identity or personal health. While data collected from individuals as the foundational research material would qualify as genetic data under the GDPR, it, absent an immediate connection to health or other PIPL-listed sensitive categories, would not apparently fall under the PIPL’s definition of sensitive personal information. This definitional approach arguably creates a narrower and more context-dependent scope for sensitive genetic data under the PIPL compared to the GDPR’s comprehensive classification, potentially leaving certain datasets outside the heightened protection regime.

An aspect of biometric data

To bridge the PIPL’s definitional gaps, China’s National Information Security Standardization Technical Committee (‘the TC260’)⁶⁷ has issued voluntary national standards regarding data protection and data security since 2020 that clarify the personally identifiable attribute of genetic data. While only remaining voluntary, these standards provide authoritative guidance for interpreting higher-level legislation like the PIPL. A general standard on personal information security

61 According to Fudan University’s statement, the publication of sample information and data involved in the study has been approved by the National Human Genetic Resources Management Department of the MOST. See ‘The First Exclusive Pangenome Reference Map for the Chinese Population’ (Fudan University News, 15 June 2023) <<https://news.fudan.edu.cn/2023/0615/c4a135371/page.htm>> accessed 8 October 2025.

62 PIPL, art 28.

63 art 28 of the PIPL provides that sensitive personal information includes information on biometric characteristics, religious beliefs, specially designated status, medical health, financial accounts, individual location tracking and the personal information of minors under the age of 14 years.

64 art 4(13) of the GDPR defines genetic data as personal data relating to the inherited or acquired genetic characteristics of a natural person which result from the analysis of a biological sample, and Recital 34 further highlights that genetic data particularly result from the chromosomal,

DNA or RNA analysis or the analysis of another element enabling equivalent information to be obtained.

65 Catherine Jasserand, ‘Legal Nature of Biometric Data: From “Generic” Personal Data to Sensitive Data: Which Changes Does the New Data Protection Framework Introduce?’ (2016) 2 European Data Protection Law Review 297; Jiajv Chen and Wei Li, ‘The Legal Status and Improvement Path of Human Genetic Data in Gene Therapy in China’ (2024) 35 Human Gene Therapy 520.

66 TJ Kasperbauer and Peter H Schwartz, ‘Genetic Data Aren’t So Special: Causes and Implications of Reidentification’ (2020) 50 Hastings Center Report 30.

67 The National Information Security Standardization Technical Committee of China (全国信息安全标准化技术委员会, ‘TC260’) is a standard-setting body which develops and oversees information security standards under the authority of the Cyberspace Administration of China (国家互联网信息办公室, ‘CAC’). See TC260, ‘Homepage’ <<https://www.tc260.org.cn/front/main.html>> accessed 8 October 2025.

lists an individual's genes alongside fingerprints and facial recognition features as examples of 'personal biometric recognition information'.⁶⁸ A subsequent standard refines this, defining 'biometric recognition information' as 'personal information obtained by technical processing of the physical, biological or behavioural characteristics of a natural person that can identify the identity of the natural person alone or in combination with other information', again citing 'an individual's genes' as a prime example.⁶⁹

More directly, a dedicated 2022 standard on 'genetic recognition data' provides a clearer indication of what types of genetic data may constitute biometric data and therefore be considered as sensitive data under the PIPL.⁷⁰ It defines 'genetic recognition data' as the data 'extracted by technical means from human genetic materials characterizing the genetic information of an individual or a population that can be directly or indirectly identified to an individual or population'.⁷¹ This standard also notes that 'genetic recognition data' is synonymous with 'genetic data'. Compared to the previous standards, two features merit emphasis regarding how the definition of genetic data is formulated in this standard. First, the scope of 'genetic (recognition) data' here covers data that can directly or indirectly identify not only an individual but also a certain group of people. In the latter case, the PIPL would no longer be applicable; instead, the Data Security Law (DSL) may take precedence if such population genetic (recognition) data triggers the threshold of 'important data' as defined by law (to be discussed in the section 'Genetic data under the Chinese data protection law'). Secondly, it cross-references the aforementioned national standard focusing on 'biometric recognition information', strongly implying that genetic (recognition) data fall within the PIPL's sensitive personal data protections.

An aspect of health data

In addition to its biometric attribute, China's national standards also situate genetic data within the category of health data under certain situations (and therefore a

category of sensitive personal information under the PIPL). The national standard *Information Security Technology—Guide for Health Data Security* lists genetic counselling data and data generated by gene sequencing as examples of 'health status data'.⁷² It further specifies that personal health data include 'data from examinations or tests of body substances, such as biological macromolecules like DNA, RNA, and proteins', as well as 'data related to medical research, such as biobank and whole genome sequencing results'.⁷³ While not its primary purpose, this standard indirectly explains the scope of genetic data, more or less, in the context of healthcare, reinforcing its status as sensitive personal information.

Taken together, it can therefore be reasonably argued that individual genetic data typically falls into the scope of sensitive personal information under China's PIPL and is interpreted as a sub-category of either biometric or health data, despite the law's silence on 'genetic data' per se. These relevant non-mandatory national standards also support this view. This approach departs from the approach of the EU's GDPR, which, by Article 9, distinguishes genetic data from biometric data and treats them as separate special categories (to be discussed in the section 'Comparative analysis: divergent definitional choices and paradigms in Chinese and EU law').⁷⁴ Moreover, the PIPL appears to leave conceptual space for certain genetic information not to be classified as sensitive where it does not reveal one's biometric characteristics or health status. Nevertheless, no authoritative court decisions or formal regulatory opinions have yet settled the classification of genetic data under the PIPL; this evidentiary gap therefore leaves the legal status of such data indeterminate for researchers and institutional actors in practice.

Under the PIPL, processing sensitive data triggers a range of heightened obligations compared to non-sensitive personal data, such as the mandatory performance of data protection impact assessments, stricter protection measures, more specific purposes for processing, and a higher threshold for processing

68 Standardization Administration of China, GB/T 35273-2020, *Information Security Technology—Personal Information Security Specification* (信息安全技术—个人信息安全规范) (issued 6 March 2020, effective 1 October 2020) art 3.2 and Annex A.

69 Standardization Administration of China, GB/T 40660-2021, *Information Security Technology—General Requirements of Biometric Information Protection* (信息安全技术—生物特征识别信息保护基本要求) (issued 12 October 2022, effective 1 May 2023), art 3.3.

70 Standardization Administration of China, GB/T 41806-2022, *Information Security Technology—Security Requirements of Genetic Recognition Data* (信息安全技术—基因识别数据安全要求) (issued 12 October 2022, effective 1 May 2023).

71 GB/T 41806-2022, art 3.2.

72 Standardization Administration of China, GB/T 39725-2020, *Information Security Technology—Guide for health data security* (信息安全技术—健康医疗数据安全指南) (issued 14 December 2020, effective 1 July 2021), art 6.1.

73 GB/T 39725-2020, Annex A.

74 art 9(1) of the GDPR lists the special categories of personal data of which the processing shall be prohibited in principle as well as the circumstances where the processing is allowed. These special categories of personal data include personal data revealing racial or ethnic origin, political opinions, religious or philosophical beliefs, or trade union membership, and the processing of genetic data, biometric data for the purpose of uniquely identifying a natural person, data concerning health or data concerning a natural person's sex life or sexual orientation.

necessity.⁷⁵ Where relying on individual consent as the legal basis, the consent should be obtained ‘separately’ for sensitive personal data, beyond the general requirements of ‘being obtained voluntarily and explicitly by the fully informed individual’.⁷⁶ However, the term of ‘separate consent’ is not clearly defined in the PIPL or explained by the regulator, which may create practical challenges for large-scale data analytics in medical and health research by reducing the volume of data available for analysis.⁷⁷ Similar to the GDPR, individuals enjoy a series of data subject rights when their personal data is processed.⁷⁸ In the case of processing sensitive data, the scope of the right to information is expanded, creating a further potential burden for processors.⁷⁹ While a granular analysis of these specific processing rules falls outside this article’s primary focus, their impact on scientific research is a topic of considerable importance that merits future investigation. Crucially, the PIPL allows other laws and administrative regulations to impose stricter requirements for the processing of sensitive personal data where they require certain government approval or impose other types of restrictions.⁸⁰ This provision is particularly salient for genetic data, which is concurrently governed by the more stringent biosecurity framework with its *ex ante* government approval procedures.

Genetic data as non-personal ‘Important Data’

As the PIPL applies only to personal data, it excludes aggregate forms of genetic data that relate to populations rather than individuals. Nevertheless, China’s national standards, as discussed in the section ‘Genetic data under the Chinese data protection law’ indicates that the term ‘genetic data’ also characterizes the genetic information of identifiable populations.⁸¹ In this sense, the concept of genetic data in Chinese law possesses a dual nature—personal and non-personal—and the latter is captured by a data-security-driven national law rather

than the PIPL. This has consequences for *inter alia* researchers involved in the analysis of genetic data because it introduces a further layer of legal complexity.

In 2021, China enacted the DSL to govern non-personal data through a risk-oriented and tiered framework targeting ‘important data’, which was first mentioned in China’s Cybersecurity Law (‘CSL’) in 2016,⁸² and ‘national core data’.⁸³ Departing from the EU-inspired PIPL, the DSL is recognized as a significant innovation in China’s data regulatory regime, creating a comprehensive architecture to assess and manage potential risks from all types of non-personal data except for state secrets.⁸⁴ The DSL establishes a data classification system based on the significance of data to economic and social development and the potential harm to national security, public interests, or the legitimate rights of individuals and organizations should data be tampered with, damaged, leaked, or illegally accessed or used.⁸⁵ Entities processing ‘important data’ must satisfy enhanced obligations under the DSL, such as reporting impact assessment results to regulators, designating a dedicated data security officer, and deploying robust risk-monitoring systems.⁸⁶

This data security framework is significant as non-personal population genetic data may be classified as ‘important data’ in certain cases. While neither the CSL nor the DSL contained a precise explanation of ‘important data’, the *Regulations on the Management of Network Data Security*, adopted in September 2024, formally and firstly define ‘important data’ as data ‘in specific fields, pertaining to specific groups or regions, or reaching a certain level of precision and scale’ which, if compromised, ‘may directly endanger national security, economic operations, social stability, public health, and safety’.⁸⁷ This statutory formulation, prefigured in previous national standards, underscores that ‘important data’ transcends enterprise-level or individual-level concerns, and supplies concrete criteria (eg fields, affected groups, precision/scale, and harm scenarios) that can guide sectoral catalogue development under the DSL.⁸⁸

75 PIPL, arts 28 and 55.

76 *ibid*, art 14.

77 Yu Yao and Fei Yang, ‘Overcoming Personal Information Protection Challenges Involving Real-World Data to Support Public Health Efforts in China’ (2023) 11 *Frontiers in Public Health* 1265050.

78 According to Ch 4 of the PIPL, data subjects have the right to information, the right to decision, the right to restriction or objection, the right to access and copy, the right to portability, the right to correction, the right to deletion, the right to explanation, and the right to remedy.

79 PIPL, arts 30 and 17.

80 PIPL 2021, art 32.

81 GB/T 41806-2022, art 3.2.

82 Cybersecurity Law of the People’s Republic of China (中华人民共和国网络安全法) (promulgated by the Standing Committee of the National

People’s Congress, 7 November 2016, effective 1 June 2017) (‘CSL’), arts 21 and 37.

83 Data Security Law of the People’s Republic of China (中华人民共和国数据安全法) (promulgated by the Standing Committee of the National People’s Congress, 10 June 2021, effective 1 August 2021) (‘DSL’).

84 Creemers (n 36).

85 DSL, art 21.

86 DSL, Chap IV.

87 Regulations on the Management of Network Data Security (网络数据安全条例) (promulgated by the State Council of China, 24 September 2024, effective 1 January 2025), art 62(4).

88 Standardization Administration of China, GB/T 43697-2024, *Data Security Technology—Rules for Data Classification and Grading* (数据安全技术—数据分类分级规则) (issued 21 March 2024, effective 1 October 2024), art 3.2.

Particularly, these standards identify the data ‘reflecting characteristics and genetic information of ethnic groups’ as a factor in identifying ‘important data’,⁸⁹ creating a clear intersection with the objectives of China’s Biosecurity Law and HGRs protection. That said, the precise alignment between the DSL and the Biosecurity Law—how the same dataset is treated when it implicates both ‘important data’ controls and HGRs management—remains to be clarified in implementing rules and inter-agency practice. Notwithstanding the DSL tasks sectoral and regional regulators with establishing detailed catalogues to delineate which data types qualify as ‘important data’ in their domains,⁹⁰ competent biosecurity authorities have yet to promulgate clear guidance on prospective classification of population-level genetic datasets and their coordinated governance under the intersecting data-security and biosecurity regimes.

The authors suggest that these criteria outlined above remain too vague for companies or researchers to self-assess whether they possess ‘important data’ and therefore what obligations they must meet. This problem is exacerbated by the slow issuance of definitive regulatory catalogues, with existing guidance remaining too disparate. A key clarification emerged from the updated cross-border data transfer rules issued by the Cyberspace Administration of China (CAC), China’s central data regulator with uniquely broad rulemaking and enforcement powers,⁹¹ in March 2024, which declares that data attain ‘important’ status only when regulators have notified the entity or publicly designated it as such.⁹² The authors then would argue that, in practice, this mechanism implies that the determination of whether certain population-level genetic datasets fall within the category is likely to depend less on statutory interpretation and more on the subsequent judgment of regulators. Such determinations may incorporate considerations linked to broader national priorities, including national security and public health, but the rationale

and process behind these decisions are not made transparent to regulated entities. As a result, the exercise of regulatory discretion introduces a layer of unpredictability that compounds the uncertainty already inherent in the broad legal definition of ‘important data’.

Further notable clarification has emerged from regional guidance in several China’s pilot free-trade zones, where local authorities have the discretion to designate the types of data should be subject to export controls.⁹³ Regulators in Beijing and Jiangsu Province, for instance, have indicated that large-scale genetic testing datasets encompassing more than 100,000 individuals qualify as ‘important data’ and are therefore subject to the CAC’s outbound security assessment requirements.⁹⁴ They also affirm that human genetic data that attain the scale or precision thresholds specified by competent national authorities will be treated as ‘important data’ and must be transferred abroad via the administrative licensing or filing routes established under the biosecurity regime, rather than the DSL’s security-assessment pathway.⁹⁵ Nevertheless, such regional guidance does not specify the numerical or technical parameters that constitute the referenced scale or precision thresholds.

It bears emphasizing that the Biosecurity Law’s requirements operate independently of and in parallel to the data security regime. That is to say, its definition of ‘HGR information’ and attendant approval procedures are not superseded by the related concepts of ‘sensitive personal information’ and ‘important data’ under China’s data protection framework. Despite the broad umbrella of ‘important data’ and the limited clarification on its relation to human genetic information, it appears that Chinese data regulators have begun to explore points of interaction between the DSL and the Biosecurity Law. However, this regulatory experimentation has so far been confined to regional initiatives, leaving its nationwide binding effect and influence on central regulatory practice an open question.

89 Annex G of GB/T 43697-2024 provides a detailed list of factors that should be considered when identifying important data, such as the overall operation of critical information infrastructure, the economic operation status of critical sectors, items subject to export control (eg major national achievements in scientific and technological innovation).

90 DSL, art 21.

91 Jamie P Horsley, ‘Behind the Facade of China’s Cyber Super-Regulator’ (*DigiChina of Stanford University*, 2022) <<https://digichina.stanford.edu/work/behind-the-facade-of-chinas-cyber-super-regulator/>> accessed 8 October 2025.

92 Provisions on Promoting and Regulating the Cross-Border Data Flows (促进和规范数据跨境流动规定) (issued by the Cyberspace Administration of China, 22 March 2024) (here after ‘Cross-Border Data Flows Provisions’), art 2.

93 Cross-Border Data Flows Provisions, art 6.

94 The People’s Government of Beijing Municipality, ‘China (Beijing) Pilot Free Trade Zone Data Outbound Management List (Negative List) (2024) (中国(北京)自由贸易试验区数据出境管理清单(负面清单)(2024版)’ (30 August 2024) <https://www.beijing.gov.cn/zhengce/zhengcefagui/202409/t20240902_3787646.html> accessed 8 October 2025; Cyberspace Administration of Jiangsu, ‘China (Jiangsu) Pilot Free Trade Zone Data Outbound Management List (Negative List) (2025) (中国(江苏)自由贸易试验区数据出境管理清单(负面清单)(2025版)’ (13 August 2025) <https://www.jszx.gov.cn/fwhd/tzgg/202508/t20250813_3568905.shtml> accessed 8 October 2025.

95 *ibid.*

Comparative analysis: divergent definitional choices and paradigms in Chinese and EU law

Foundational paradigms: no equivalent biosecurity regime in the EU law

The most profound divergence between the Chinese and EU approaches to defining genetic data lies not in specific rules, but in their foundational legal philosophies and paradigms. As the section ‘China’s dual frameworks governing the concept of genetic data: national sovereignty versus individual rights’ discussed, China’s framework is uniquely characterized by a dual focus on national sovereignty and individual data interests. The former is primarily embodied by its Biosecurity Law, which stands apart from any single EU instrument and has no direct equivalent in EU law. The approach in Europe remains focused on respecting individual autonomy, typically through obtaining informed consent from the individuals concerned.⁹⁶

A law anchored in the concept of nation and ethnicity?

China’s Biosecurity Law situates HGRs within a state-centric biosecurity context, treating the genetic makeup of its population as a strategic national asset.⁹⁷ As discussed in the section ‘China’s dual frameworks governing the concept of genetic data: national sovereignty versus individual rights’, the underlying motivation of this approach could be traced back to the history of safeguarding the collective genetic property of the nation from perceived threats of foreign exploitation or ‘biopiracy’. Its provisions are unsurprisingly framed in territorial and national-based terms. For instance, Article 56 of the Biosecurity Law prohibits foreign organizations from collecting or preserving the country’s HGRs within the territory of China, and from providing China’s HGRs abroad.⁹⁸ On that statutory reading, the extraterritorial collection of genetic material from persons of Chinese descent (regardless of whether they are Chinese nationals)—when that

collection occurs wholly outside China and does not involve activities on Chinese soil—is unlikely, as a matter of statutory scope, to fall within China’s biosecurity regime.

In contrast, the GDPR adopts a data-centric approach, which considers any personal genetic information merits protection irrespective of ethnicity, and the Regulation’s applicability is determined by processing location and connections to EU residents (whenever they are Europeans, Americans, or Chinese).⁹⁹ It is also not linked to membership of particular ethnicities, in stark contrast to the Chinese law’s apparent national and ethnic linkage. This latter point prompts an interesting extraterritorial scenario where health-related research is performed on people of Chinese ethnic origin that live outside of China, even if they lack Chinese citizenship. In practice, for example, pharmaceutical companies based in Europe would involve participants of Chinese descent in clinical trials to obtain key data from different ethnic origins for medicinal product R&D.¹⁰⁰ This case will not be captured by China’s biosecurity regime but will be governed by EU regulatory frameworks, notably the GDPR and the Clinical Trials Regulation (Regulation (EU) No 536/2014).¹⁰¹

A law that relates not only to human tissues, but also to the information derived from them

A further point of significant divergence lies in the notion of ‘HGRs’. Under the Chinese Biosecurity Law (see the section ‘Genetic data as a national resource: the wide reach under the biosecurity law’), this notion encompasses not only human genetic data but also a range of physical tissues from which such data could be derived. These tissues, termed ‘HGR materials’, represent human tissues and other biological materials that could be the subject of genetic sequencing.¹⁰² In contrast, the EU’s GDPR, as the primary framework governing genetic data, governs only the personal data produced from such tissues, not the tissues themselves.¹⁰³ While the EU and its Member States have developed a range of laws regulating the use of human

96 Jane Kaye and others, ‘Consent for Biobanking: The Legal Frameworks of Countries in the BioSHaRE-EU Project’ (2016) 14 *Biopreservation and Biobanking* 195.

97 McKibbin and Shabani (n 31).

98 PRC Biosecurity Law, art 56.

99 art 3(1) of the GDPR establishes the application to any processing of personal data by controllers and processors established in the EU, regardless of where processing takes place. art 4(1) defines personal data as any information relating to an identified or identifiable natural person, without distinguishing nationalities or ethnicities.

100 eg Pfizer’s Clinical Research Unit Brussels announced to conduct a clinical study with participants of Chinese descent, regardless of nationality and residence, for evaluating an investigational medicinal product. See

Pfizer, ‘Healthy Chinese participants needed’ <<https://www.pfizerclinicaltrials.com/pcru-brussels/healthy-chinese-participants>> accessed 9 October 2025.

101 Regulation (EU) No 536/2014 of the European Parliament and of the Council of 16 April 2014 on clinical trials on medicinal products for human use, and repealing Directive 2001/20/EC (‘Clinical Trials Regulation’), OJ 2014 L 158/1.

102 PRC Biosecurity Law, art 85(8); RMHGR, art 2; RMHGR Implementation Rules, art 2.

103 Mark Taylor, ‘Privacy’ in Mark Taylor (ed), *Genetic Data and the Law: A Critical Perspective on Privacy Protection* (CUP 2012) <<https://www.cambridge.org/core/product/43B7D037DDE45A2A0672E9CEEDE67A96>> accessed 8 October 2025.

tissues, these instruments typically derive their scope from a specific application, for example, biobanking, blood donation, or clinical trials, rather than from the mere potential to extract genetic data.

At the EU level, for example, the Tissues and Cells Directive (Directive 2004/23/EC) applies to ‘*the donation, procurement, testing, processing, preservation, storage and distribution of human tissues and cells intended for human applications*’.¹⁰⁴ Its rationale is to address specific physical risks of disease transmission or other serious harms arising from clinical use, making it applicable only where tissues are intended for direct medical treatments or applications on the human body.¹⁰⁵ Similarly, there are other specific national legal frameworks that apply to the use of human tissues in specific areas, such as Germany’s Transplantation Act (Transplantationsgesetz, TPG).¹⁰⁶ It imposes specialized rules on the removal, storage, and transplantation of human organs and tissues.¹⁰⁷ Given the charged ethical context (involving inter alia end-of-life care) and the risk of other harms such as the transmission of disease, individual autonomy, and dignity, a specialized legal framework is deemed warranted.

Even for national legislation that is capable of a broader application, the focus is usually arguably narrower than China’s Biosecurity Law. For instance, the UK’s Human Tissue Act 2004 (HTA) adopts a wider material scope, covering the removal, storage, use, and disposal of human tissue for various purposes such as medical research, education, organ transplantation, and post-mortem examinations.¹⁰⁸ Yet even here, the framework remains strictly purpose-bound—even the broad term ‘medical research’ is defined in a relatively limited way.¹⁰⁹ Notably, its underlying rationale is to protect the interests of individual donors rather than to safeguard ‘British’ genetic tissue or data as a collective national asset.¹¹⁰ Likewise, the German Genetic Diagnostics Act (GenDG)¹¹¹ oversees the processes of genetic testing and genetic information handling,

guaranteeing that biobanks manage genetic data responsibly while protecting individuals from genetic discrimination in areas like insurance and employment inter alia by ensuring their right to self-determination regarding their genetic information.¹¹²

Taken together, these European frameworks illustrate the multi-level and fragmented regulatory environment, operating across both Union-level and national instruments. In contrast, China’s Biosecurity Law applies by virtue of the mere possession of ‘HGR materials’, such as organs, tissues, or cells, from which genetic information of the Chinese population and ethnic groups could be derived. Under this law, such materials, alongside genetic data, are regarded as national assets. This dual distinction—both in the breadth of coverage and the underlying rationale—marks a fundamental departure from European approaches. Whereas EU instruments are primarily inspired by the need to protect individuals (eg donors), China’s is premised on safeguarding the collective genetic property of the nation. European frameworks, even when invoking broader societal interests, are not predicated on a desire to protect national assets and therefore do not construct comparable barriers to the transfer or utilization of biological materials based on the concepts of nationality or national ownership.

While regulations like the EU’s on access to genetic resources and benefit-sharing (Regulation (EU) No 511/2014) exist, it expressly excludes HGRs from its regulatory scope and does not advance biosecurity aims per se.¹¹³ The European Health Data Space (EHDS) Regulation, a more recent component of the Union’s data strategy, aims to enhance data sovereignty in the health sector by establishing harmonized rules on data access, interoperability, reuse and cross-border exchange of health-related datasets.¹¹⁴ While genetic data are clearly covered by the EHDS, it follows the formal definition as provided in the GDPR and does not extend its inherent meaning.¹¹⁵ Notwithstanding the regulatory

104 Directive 2004/23/EC of the European Parliament and of the Council of 31 March 2004 on setting standards of quality and safety for the donation, procurement, testing, processing, preservation, storage, and distribution of human tissues and cells, OJ 2004 L 102, art 2.

105 V Weisbach, ‘Regulation of Cell-Based Medicine—the German Experience’ (2010) 5 ISBT Science Series 252.

106 Gesetz über die Spende, Entnahme und Übertragung von Organen und Geweben (Transplantationsgesetz—TPG) (Transplantation Act) of 5 November 1997, Bundesgesetzblatt (BGBl.) I, 2631.

107 Wolf Rommel and Hartmut HJ Schmidt, ‘Organ Transplantation in Germany—Legal Framework and Organizational Management’ (2010) 45 Anesthesiologie, Intensivmedizin, Notfallmedizin, Schmerztherapie: AINS 348.

108 Human Tissue Act 2004, c 30.

109 Schedule 1, s 5 of the act actually refers to ‘research in connection with disorders, or the functioning, of the human body’.

110 MDD Bell, ‘The UK Human Tissue Act and Consent: Surrendering a Fundamental Principle to Transplantation Needs?’ (2006) 32 Journal of Medical Ethics 283.

111 Gesetz über genetische Untersuchungen bei Menschen (Gendiagnostikgesetz—GenDG) (Genetic Diagnostics Act) of 31 July 2009, Bundesgesetzblatt (BGBl.) I, 2529.

112 GenDG, s 1.

113 Regulation (EU) No 511/2014 of the European Parliament and of the Council of 16 April 2014 on compliance measures for users from the Nagoya Protocol on Access to Genetic Resources and the Fair and Equitable Sharing of Benefits Arising from their Utilization in the Union, OJ 2014 L 150.

114 European Commission, ‘European Health Data Space Regulation (EHDS)’ (European Commission) <https://health.ec.europa.eu/health-digital-health-and-care/european-health-data-space-regulation-ehds_en> accessed 8 October 2025.

115 EHDS Regulation, art 2(1).

significance of these EU instruments for the governance of biological and genetic data, their sectoral aims, institutional architecture, and normative priorities do not together constitute a single, consolidated statutory framework comparable in scope or biosecurity-orientation to China's Biosecurity Law.

Genetic data are implicitly recognized as sensitive data in China whereas explicitly in Europe

One of the most noticeable differences between the PIPL and the GDPR is that in the former, genetic data is not explicitly invoked as being sensitive data, whereas in the GDPR it is.¹¹⁶ Article 4(13) of the GDPR defines 'genetic data' as '*personal data relating to the inherited or acquired genetic characteristics of a natural person which give unique information about the physiology or the health of that natural person and which result, in particular, from an analysis of a biological sample from the natural person in question*'. Crucially, the GDPR explicitly designates genetic data as 'special categories of personal data', placing it alongside one's biometric data and data revealing health.¹¹⁷ This per se classification means that a number of extra burdens foreseen in the GDPR for those processing genetic data will always apply, notably the prohibition of processing genetic data unless a specific exemption applies.¹¹⁸ This explicit and uniform protection is a cornerstone of the EU framework, providing a high level of certainty that all personal genetic data, regardless of the context or purpose of its processing, is subject to the strictest safeguards.

This, however, is not the case with the PIPL as it adopts an implicit and context-dependent approach, rather than expressly enumerating genetic data within the scope of sensitive personal data. As discussed in the section 'Genetic data under the Chinese data protection law', human genetic data would in many cases be considered sensitive when it falls under the meaning of 'health data' or 'biometric data', two listed categories of sensitive personal data in the PIPL's context. While this captures a vast amount of genetic data used in clinical and research settings, it creates a potential lacuna. For instance, genetic data used in research that does not have immediate health or identification goals—such as investigations into human palaeontology and migration

patterns—would not, on its face, fall under the PIPL's sensitive categories.¹¹⁹ Such research, while revealing information about certain genetic sequences, does not usually reveal information concerning human health. Accordingly, it could be argued that the PIPL's approach to defining genetic data creates a more contingent, use-specific (ie for identification or health-related purposes), and potentially narrower scope of protection compared to the GDPR's uniform standard, regardless of the context.

Interestingly, one might also wonder about the extent to which raw genetic data (ie in the absence of interpretive analysis) could therefore be considered sensitive data. The answer to this question would most likely depend on the size of the genetic sequence in question. Large-scale datasets, including but not limited to genome-wide samples (GWAS), could readily yield health inferences once combined with readily available knowledge, contemporary computational, and algorithmic tools.¹²⁰ It is also likely to evolve over time, given that increases in computing and algorithmic power, together with the availability of ever more complementary data, may mean that such analyses become increasingly easy to perform.¹²¹ This changing reality could, in theory, cause uncertainty for researchers about what forms of genetic data will indeed be classified as sensitive data or not under data protection law.

Divergent definitional choices to convergent outcomes: the legal uncertainty of genetic data as biometric data under the PIPL

As analysed in the section 'Genetic data as a national resource: the wide reach under the biosecurity law', genetic data, by its very nature, are in certain contexts classified as biometric data under China's data protection framework. This also depends largely on how biometric data are defined in law.¹²² Under the GDPR, the definition of biometric data seems to implicitly see a role for intent, stating that biometric data are '*personal data resulting from specific technical processing relating to the physical, physiological or behavioural characteristics of a natural person, which allow or confirm the unique identification of that natural person, such as facial images or dactyloscopy data*'.¹²³ By invoking the need for 'specific technical processing' and 'unique

116 For a more in-depth discussion on genetic and other forms of sensitive data under the GDPR, see Paul Quinn and Gianclaudio Malgieri, 'The Difficulty of Defining Sensitive Data—The Concept of Sensitive Data in the EU Data Protection Framework' (2021) 22 German Law Journal 1583.

117 GDPR, art 9(1).

118 GDPR, art 9(2).

119 Quinn and Quinn (n 48).

120 Carla G van El and others, 'Whole-Genome Sequencing in Health Care' (2013) 21 European Journal of Human Genetics 580.

121 Quinn and Quinn (n 48).

122 Els J Kindt, 'Biometric Data, Data Protection and the Right to Privacy' in Els J Kindt (ed), *Privacy and Data Protection Issues of Biometric Applications: A Comparative Legal Analysis* (Springer 2013) <https://doi.org/10.1007/978-94-007-7522-0_3> accessed 8 October 2025.

123 GDPR, art 4(14).

identification', the GDPR seemingly aims to confine biometric data to that have undergone deliberate procedures expressly designed to enable personal identification. Given that such forms of processing are unlikely to be carried out unintentionally, such a requirement also implicitly invokes an important element of intent.¹²⁴ Accordingly, the EU seemingly does not subscribe to the idea that data can be 'incidentally biometric'. Given this, it is very unlikely that routine processing of genetic data without an identification intention would be classified as biometric data processing in the absence of specific technical processing to allow the 'unique identification' of a specific individual.

The Chinese PIPL, in contrast, appears to operate differently. While the PIPL itself does not define biometric data, influential national standards as the section 'Genetic data under the Chinese data protection law' discusses (ie GB/T 40660-2021) define that 'biometric identification data' is generated from biological features 'using technical means'. At first glance, this might seem similar to the GDPR's 'technical processing' element. However, the context is crucially different. The definition of 'biometric identification data' provided in standards describes it as those 'can identify the identity of the natural person alone or in combination with other information', highlighting the capability of identification rather than the intent of processing. This seemingly indicates that whether data qualify as biometric under the PIPL depends on its inherent identifiability, namely, the ease with which a unique individual can be biometrically discerned, rather than its specific use case. Under such a paradigm, data that would require less additional processing to be used to identify unique individuals would be more likely to be classed as biometric and thus sensitive in nature. One can therefore surmise that large genetic sequences, including whole-genome sequences (WGS), are more likely to be considered biometric data given that they significantly enhance the capability of individual identification.¹²⁵ Furthermore, the determination of whether such data is of a biometric nature may also become dynamic over time. As computing power grows and potentially complementary datasets proliferate, such data may more readily qualify as biometric—for example, even modest genetic fragments could suffice for identification.¹²⁶

This can be contrasted with the situation in the EU's GDPR where, in the absence of some additional processing techniques, genetic data remains outside the

biometric category and is in principle treated as a separate category of sensitive personal data. The result of this difference is that, although the PIPL does not explicitly genetic data as a sensitive category, the de facto outcome for genetic data samples of a large size is often the same: they will likely be considered biometric by virtue of their intrinsic identifiability and thus sensitive data. However, the legal reasoning to reach this conclusion creates significant divergence. While the destination may be similar (treating genetic data as sensitive), the journey under Chinese law introduces far more legal indeterminacy. For researchers, this uncertainty means that they must continuously assess whether novel sequencing methods or data-linkages convert seemingly innocuous fragments into regulated personally biometric (in this context, genetically identifiable) data. This shifting compliance threshold may have a negative effect on scientific research with genetic data, complicating compliance, ethical review, and cross-border collaboration.

Beyond individuals: genetic data as 'Important Data' in Chinese law

A further critical divergence in the definitional choices of genetic data between these two jurisdictions is China's regulation of non-personal, population-level genetic datasets through the DSL. As the section 'Comparative analysis: divergent definitional choices and paradigms in Chinese and EU law' discusses, this introduces a category of regulated data—'important data'—that has no direct parallel in the individual-centric GDPR and other data-related legal instruments of the EU.

The DSL governs data based on its potential harm to national security, economic operations, and public health. 'Important data' is broadly defined as data that, if compromised, could endanger these national interests. While the definition remains relatively vague, national standards and regional regulatory guidance have indicated that data reflecting the 'genetic information of ethnic groups' and large-scale genetic testing datasets (eg those with over 100,000 individuals) are likely to qualify. The DSL poses heightened and additional security obligations to those processing 'important data', which for researchers translates into significant practical hurdles. For instance, mandatory risk assessments and regulatory reporting obligations add a substantial compliance layer to scientific projects, and the strict cross-

124 Quinn and Malgieri (n 116).

125 Holly K Tabor and others, 'Genomics Really Gets Personal: How Exome and Whole Genome Sequencing Challenge the Ethical Framework of Human Genetics Research' (2011) 155A American Journal of Medical

Genetics Part A 2916 <<https://doi.org/10.1002/ajmg.a.34357>> accessed 8 October 2025.

126 Quinn and Quinn (n 48).

border transfer controls on ‘important data’ can severely complicate data sharing in international collaborations and the publication of findings in global repositories.¹²⁷ This means that even when a genetic dataset is fully anonymised and falls outside the PIPL’s scope, it may still be subject to stringent state control if it is deemed significant to broader national interests from the DSL’s view. To some extent, this additional regulatory layer also echoes the state-centric and security-oriented paradigm of the Biosecurity Law, underscoring China’s treatment of population-level genetic information as a matter of national strategic importance, a concept absent from the EU’s existing framework.

Structural divergences of legal systems also shape the understanding of what genetic data is

Finally, the distinctive legal order of each jurisdiction also contributes to shaping each’s definitional choice of genetic data, which impacts legal certainty and compliance. Fundamentally, the EU law system, despite its known complexity, comprises primary legislation as EU treaties, secondary legislation as regulations, directives, and decisions and an authoritative body of the Court Justice of European Union (CJEU) jurisprudence.¹²⁸ It does not, however, feature a formally tiered and hierarchical legal architecture as is the case in China (see the section ‘The Chinese legal system: a distinct context for genetic data regulation’).¹²⁹ While EU law and national law are analytically distinct, it would be misleading to portray Member State law as simply hierarchically inferior to EU law in the field of data protection; rather, the two operate in a complementary fashion. Although in many regards EU data protection law binds Member States, each state retains sovereignty and meaningful discretion in certain areas.¹³⁰ Furthermore, the EU law system is complemented by interpretive instruments such as opinions and guidelines issued by the European Data Protection Board (EDPB).¹³¹ Those soft-law instruments carry significant practical weight and may

be considered as insightful and even authoritative (particularly in academic writings), they are not legally binding.¹³²

Compared with the Chinese legal system, it can be observed that EU law lacks an equivalent to China’s binding administrative regulations and departmental rules. As discussed in the section ‘The Chinese legal system: a distinct context for genetic data regulation’, administrative regulations occupy a lower hierarchical position than legislation enacted by the NPC (such as the PIPL, the DSL, and the Biosecurity Law), while remaining legally binding, having a narrower applicable scope, and being adopted through a comparatively less cumbersome and faster legislative procedure. Those instruments must be applied by courts and administrative organs unless they conflict with higher-ranking legislation.¹³³ For legal issues concerning genetic data, these subordinate instruments play an important role in shaping the definition and operational requirements of genetic data in Chinese law, both within the data-protection regime and the biosecurity regime.

These structural differences have mixed consequences for researchers working with genetic material. On one view, the availability of binding, domain-specific rules (eg as is the case with China’s administrative regulations) can provide clearer and sector-tailored obligations for practitioners in fast-moving technical fields such as genomics. Administrative regulations in China can be promulgated to meet sectoral needs more quickly than primary legislation while retaining legal force. Meanwhile, it should be noted that the legal force and sector-specific target of administrative regulations do not necessarily eliminate doctrinal indeterminacy. As previous sections demonstrate, statutory terms such as ‘HGRs’ and ‘important data’ can be formulated broadly in primary law and further specified by subordinate binding rules, producing both procedural clarity in certain domains and substantive breadth that can generate uncertainty. In contrast, one could also make the argument that the inherently non-binding nature of European guidelines affords regulators and relevant

127 Reuters, ‘China’s Data Protection Rules Prompt Pause from Major European Research Funders’ (*The Economic Times*, 25 April 2025) <<https://economictimes.indiatimes.com/tech/technology/chinas-data-protection-rules-prompt-pause-from-major-european-research-funders/articleshow/120603659.cms>> accessed 8 October 2025.

128 European Commission, ‘Types of EU Law’ <https://commission.europa.eu/law/law-making-process/types-eu-law_en> accessed 8 October 2025.

129 Ignazio Castellucci, ‘Rule of Law with Chinese Characteristics’ (2007) 13 Annual Survey of International & Comparative Law 19 <<https://digitalcommons.law.ggu.edu/annlsurvey/vol13/iss1/4>> accessed 9 October 2025.

130 art 9(4) of the GDPR eg allows Member States to set further rules concerning the processing of health, genetic, or biometric data. For more see Quinn and Malgieri (n 116). In the area of policing, EU’s justice and

national security competences are lesser than in other area of data protection with Member States having greater space to define their own laws. See also Monika Zalnieriute, ‘A Struggle for Competence: National Security, Surveillance and the Scope of EU Law at the Court of Justice of European Union’ (2022) 85 *The Modern Law Review* 198.

131 European Data Protection Board, ‘Tasks and Duties’ <https://www.edpb.europa.eu/about-edpb/what-we-do/tasks-and-duties_en> accessed 8 October 2025.

132 Miroslava Scholten (ed), *Research Handbook on the Enforcement of EU Law* (Edward Elgar Publishing 2023) <<https://www.elgaronline.com/edcollbook/book/9781802208030/9781802208030.xml>> accessed 9 October 2025.

133 PRC Legislation Law, art 108(2).

institutions greater adaptiveness and iterative interpretive space in response to rapidly changing social and technical contexts than binding law can.¹³⁴ Those guiding instruments can be updated or withdrawn with greater ease to allow them to follow various legal, scientific, societal, and technical evolutions in fast-evolving field of genomics, whereas legally binding rules must be reconciled with higher-ranking law before amendment.

Conclusion

A comparative analysis of the legal frameworks defining genetic data in China and the EU reveals that the divergences between them extend beyond mere textual differences to encompass foundational legal philosophies, regulatory structures, and conceptual paradigms. The definitional choices made by each jurisdiction are not arbitrary but are the logical consequence of fundamentally different underlying objectives. By mapping these differences, we can better understand their profound practical implications for genomic research, particularly for international collaborations.

The central finding is that the Chinese and EU frameworks for genetic data are anchored in two fundamentally divergent legislative philosophies: the safeguarding of national interests versus the prioritization of individual rights. This core distinction is the master key to understanding all other divergences surrounding the definitions of genetic data. The EU's approach, epitomized by the GDPR, is grounded in a rights-based tradition where genetic data is defined and protected as a category of sensitive personal data because of its intrinsic link to the dignity, autonomy, and privacy of the individual from whom it originates. This philosophy manifests in specific definitional choices: an explicit, *per se* classification of all personal genetic data as a 'special category', a clear legal distinction between genetic and biometric data, and a definition of biometric data that hinges on processing intent. The entire regulatory apparatus is designed to protect the person. In stark contrast, China's approach is principally state-centric and features dual parallel frameworks governing genetic data, that is, biosecurity and data protection. Key findings from this article's comparison show this results in: a unified governance of both physical tissues and derived data under the Biosecurity Law's expansive concept of 'HGRs'; an implicit, context-dependent definitional approach to genetic data as sensitive personal data under the PIPL, where its status is contingent on its inherent

identifiability as biometric or health data, rather than intent; and the creation of a unique category of non-personal 'important data' to control certain population-level genetic datasets under the DSL.

For international researchers engaging with genetic data, particularly in collaborations with China, the practical consequences of these findings are significant and demand a substantial shift in the multi-jurisdictional compliance mindset. First, engaging with Chinese genetic data require navigating a dual-track regulatory system of biosecurity and data protection (security), rather than assuming that compliance with the PIPL, the GDPR's closest analogue, is sufficient. Researchers must simultaneously address the entirely separate and more stringent requirements of the biosecurity regime, which governs both physical HGR materials and genetic data as national resources with an attribute of sovereignty. Anonymization, often a key risk-mitigation strategy under the GDPR, is not a safe harbour as China's approach captures both personal genetic data and population-level genetic datasets.

Secondly, the journey to a compliance outcome in China is fraught with greater legal indeterminacy. While the GDPR's explicit definition of genetic data provides a clear and uniform standard, the PIPL's implicit approach forces a continuous and context-specific assessment of whether genetic data qualifies as biometric or health data (as two categories of sensitive personal data). The definitional ambiguity of the 'important data' category under the DSL further exacerbates this uncertainty, effectively shifting the basis for classification from statutory interpretation to the discretionary and often opaque judgment of regulatory authorities. For those international researchers accustomed to the GDPR-style framework and detailed interpretive guidance, this introduces a significant layer of unpredictability that complicates risk assessment, ethical review, and project planning.

Thirdly, both legal systems possess mechanisms for adapting to technological change in genomics field, though their binding effect and flexibility differ. China's capacity to issue legally binding administrative regulations can, on one hand, provide practitioners with clearer sector-specific legal obligations more rapidly than amending primary legislation—a potential advantage in a fast-moving field like genomics. On the other hand, this procedural clarity does not eliminate doctrinal indeterminacy (eg the broadly formulated statutory terms like 'HGRs' and 'important data'). In contrast, the EU's

134 PG Casali and M Vyas, 'Data Protection and Research in the European Union: A Major Step Forward, with a Step Back' (2021) 32 *Annals of Oncology* 15.

reliance on non-binding guidance from bodies like the EDPB affords greater adaptiveness and iterative interpretive space. While these instruments lack formal legal force, they can be updated with greater ease to reflect scientific and societal evolutions, offering a flexibility that is often crucial in navigating the complexities of genomic research.

In conclusion, the definitional choices for genetic data in Chinese and EU law are not simply matters of technical legal drafting; they are also reflections of deep-seated priorities. Achieving successful and ethical Sino-EU scientific collaboration in the genomic era necessitates that stakeholders cultivate a nuanced compliance literacy extending beyond the conventional boundaries of data protection. It requires understanding and engaging with a parallel legal universe where genetic data is

conceptualized not just as individual data privacy to be protected, but as an essential national resource to be secured. Bridging this conceptual divide is the foremost challenge and the most critical step forward.

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